

# RLVR as Distributional Shrinkage

Bounded verifier information drives policy updates; rollout action entropy shrinks into problem-conditioned seeds; cross-prompt transfer reveals their geometry.

40×40

source × target matrix

608/1600

positive transfer cells

42

significant cells

## Core Hypothesis

Before RL: broad rollout/action distribution  $\pi_0(y \mid x_i)$

After RL: low-entropy problem seed  $s_i$

Between seeds: transfer geometry  $T_{ij} = p_j(\text{after } i) - p_j(\text{base})$

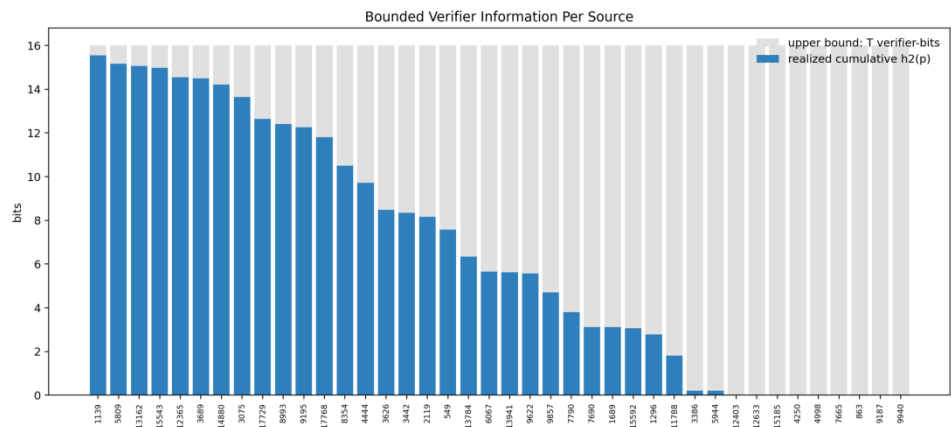
Shrink is measured by action entropy; the seed diagnostic conditions it on successful rollouts.

## 1. Bounded Information

Finite verifier signal gives a per-step upper bound:

$$H_i(t) = h_2(p_i(t)) \leq 1 \text{ bit} \quad A_i = \sum_t H_i(t)$$

Observed cumulative verifier uncertainty: 265.5 / 640 step-bits (41.5% of bound).

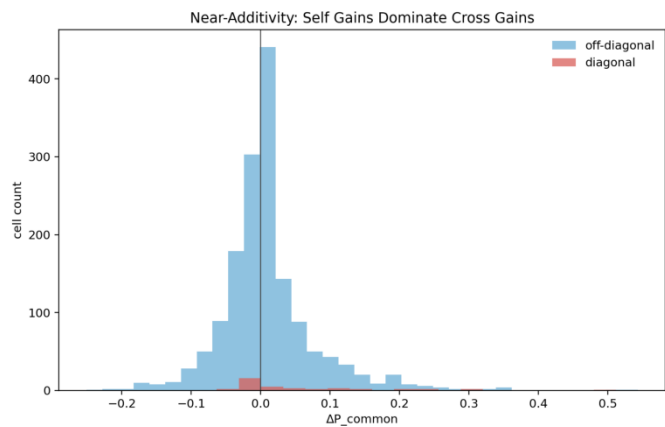


## 2. Dataset Is Mostly Additive

Self-transfer dominates cross-transfer:

$$\text{diag } \Delta P = 0.0706 \quad \text{offdiag } \Delta P = 0.0091$$

offdiag / diag = 0.129; additivity score = 0.871.

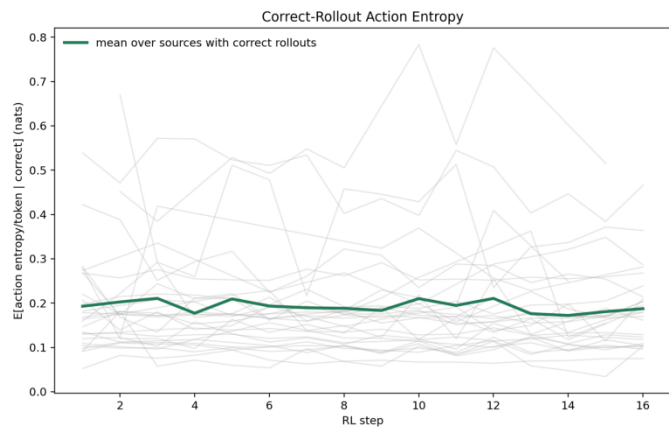


## 3. Rollout Entropy Shrinks

Successful trajectories become sharper under RL:

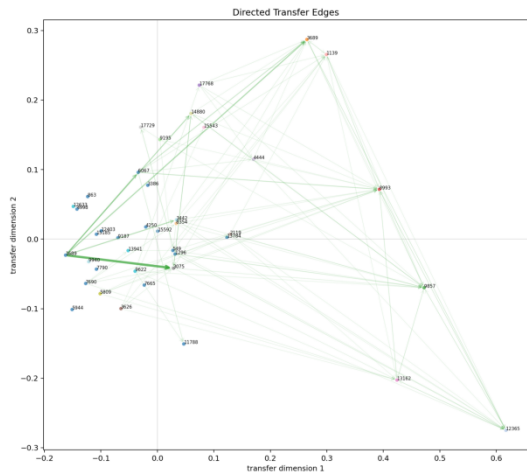
$$H_{\text{action}} \mid \text{correct: } 0.217 \rightarrow 0.202 \text{ nats/token}$$

Correct-only shrink = 0.0151; all-rollout shrink = 0.0128.



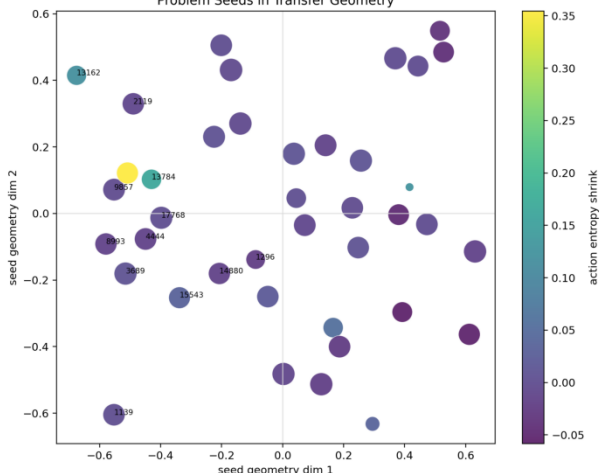
## Transfer Geometry

Directed graph: train source  $i$ , evaluate target  $j$ .



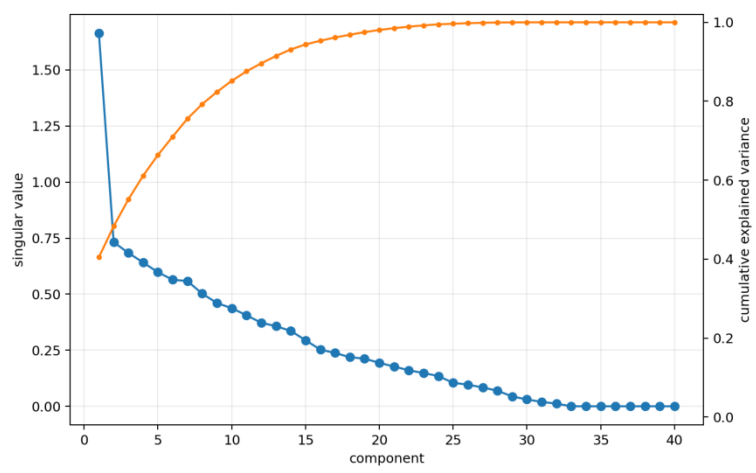
## Seed Space

Problem seeds embedded by transfer fingerprints: color shows action-entropy shrink.



## Low-Rank Structure

Transfer matrix is low-dimensional: 2D EV = 0.483, 16D EV = 0.953.



Takeaway: RLVR turns broad rollout distributions into a constellation of low-entropy problem seeds. Most dataset mass is additive; sparse cross-transfer defines the seed geometry.